Layer-Aware Native Quantization for Machine Translation: A WMT26 English-to-Chinese Submission

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Abstract

This paper describes the alonso submission to the constrained English-to-Simplified-Chinese track of the WMT26 Model Compression shared task. Starting from Gemma 3 12B, we apply native BitsAndBytes NF4 with double quantization to the 144 gate, up, and down projections in the model's 48 text-transformer MLP blocks. Attention projections, embeddings, normalization layers, the tied language-model head, and vision modules remain in BF16. The method requires no calibration or fine-tuning data and produces a mixed checkpoint whose quantized weights remain low-bit at inference. On the official 945-segment blind set, the system scores 0.7669 CometKiwi-XXL and 2.7836 MetricX-24-XXL, slightly improving on the organizer global-q4 baseline's 0.7665 and 2.8147, respectively. Its 11.8 GB artifact is 48.4% of BF16 size, and H100 throughput is essentially tied with global q4 (866.9 versus 867.9 source characters/s). On 332 reference-backed WMT25 segments, its chrF and BLEU gains over global q4 are significant under paired bootstrap resampling. A transferred selected-layer q3+Zstd codec reaches statistically indistinguishable local quality but expands to BF16-like memory after decoding.

1 Introduction

Large multilingual language models are capable machine translation (MT) systems, but their storage, accelerator-memory, and decoding requirements complicate deployment. Model compression reduces these requirements, yet translation quality can respond unevenly to the location and strength of compression (Mohammadshahi et al., 2022). The WMT Model Compression task makes this trade-off explicit by jointly evaluating quality, model size and VRAM, and inference speed (Gaido et al., 2025).

This work asks whether a layer-selection hypothesis previously studied for speech translation transfers to text MT. In an IWSLT 2026 English-to-Chinese system, applying lossy quantization and Zstandard coding only to transformer MLP projections preserved quality better than applying one codec globally (Palomino, 2026). That method primarily reduced stored bytes: selected weights had to be reconstructed into ordinary tensors for execution, so the compressed representation did not directly translate into accelerator-memory savings.

For WMT26, we retain the high-level layer-aware policy but change the representation used at inference. The submitted system selectively converts the text transformer's `gate_proj`, `up_proj`, and `down_proj` modules to native 4-bit NormalFloat (NF4) modules with double quantization (Dettmers et al., 2023). Other modules remain in BF16. This design targets 144 matrices containing 8.49 billion weight scalars while protecting attention and output-critical components. The low-bit modules are serialized and executed through BitsAndBytes, so their storage reduction is retained in GPU memory.

We make three contributions:

- a directly runnable mixed-precision Gemma 3 12B checkpoint that satisfies the WMT26 submission contract;
- a controlled local comparison with the organizer BF16 and global-q4 baselines, and with the previous selected-layer q3+Zstd method; and
- a deployment-oriented analysis that separates serialized size, post-load VRAM, generation speed, and translation quality.

The main result is a trade-off rather than dominance on every dimension. Locally, selective native q4 improves significantly over the smaller organizer global-q4 baseline by 0.63 chrF and 0.56 BLEU, but uses 2.60 GiB more peak VRAM. The official blind evaluation confirms a small neural-metric advantage over global q4 at nearly identical H100 throughput, while BF16 remains slightly better in both official quality and throughput.

2 Task and Experimental Setting

2.1 WMT26 constrained track

The WMT26 Model Compression shared task evaluates compressed systems along three dimensions: translation quality, model storage and VRAM, and decoding speed. The constrained track fixes `google/gemma-3-12b-it` as the source model (Gemma Team, 2025). Systems must remain directly derived from that model. The task covers Czech–German, English–Simplified Chinese, and English–Egyptian Arabic, with separate rankings by direction. Organizer evaluation uses a single NVIDIA H100 GPU with up to 80 GiB on Ubuntu 24.04.[1]

We submit only English–Simplified Chinese (`eng-zho_Hans`) in the constrained track. The primary system ID is `layeraware-native-mlp-q4`. No calibration, fine-tuning, or other training data is used. The submitted repository includes the model, setup instructions, and an inference entry point that accepts line-oriented source text under the organizer contract.[2]

2.2 Local development data

Local comparisons use the reference-backed WMT25 English–Chinese data distributed through the WMT26 organizer repository. The prepared set contains 332 source–reference pairs. We use these past-test references only for evaluation; they are not used to select calibration examples, tune weights, or train adapters. We distinguish these development scores from the official WMT26 blind-test results reported in Section 4.3.

2.3 Official blind evaluation

The organizer evaluated 945 English–Chinese source segments without references using CometKiwi-XXL (higher is better) (Rei et al., 2023) and MetricX-24-XXL (lower is better) (Juraska et al., 2024). Evaluation ran one system per NVIDIA H100 80 GB GPU. The released throughput is source characters divided by wall time at the largest evaluated batch; the released peak-memory field is host RSS, not GPU VRAM.[3]

2.4 Inference and metrics

Every system receives the same organizer translation prompt: "Translate the following text from English to Chinese. Return only the translation, with no explanation, labels, or quotes." The Gemma chat template is applied. Inputs are length-sorted for batching, then restored to original order. Decoding is greedy (no sampling, one beam) with batch size 8. The generation budget is the smaller of 1,024 tokens and the longest source length in the batch plus 64 tokens. Each run produces one line per source line.

We report lowercased chrF2 (Popović, 2015) and lowercased BLEU with SacreBLEU 2.5.1 and its Chinese tokenizer (Post, 2018). Statistical comparisons use SacreBLEU paired bootstrap resampling with 10,000 trials, fixed seed 12345 (Koehn, 2004). The complete paired-test signatures are `nrefs:1|bs:10000|seed:12345|case:lc|eff:no|tok:zh|smooth:exp|version:2.5.1` for BLEU and `nrefs:1|bs:10000|seed:12345|case:lc|eff:yes|nc:6|nw:0|space:no|version:2.5.1` for chrF2.

Local performance is measured on one NVIDIA RTX A6000 with 48 GiB. Each system is loaded in a fresh process. Generation time excludes processor and model loading; VRAM is the maximum observed by one-second `nvidia-smi` polling over the run. Artifact size is the serialized model payload, not the compressed submission ZIP. Because WMT26 uses a different H100 environment, local speed and memory values are diagnostic rather than official.

3 Compression Systems

All systems use the same Gemma 3 12B revision and organizer inference wrapper. Table 1 summarizes the precision policies.

3.1 Organizer baselines

The BF16 baseline is the original `google/gemma-3-12b-it` artifact. The organizer global-q4 baseline loads eligible

[1] `https://www2.statmt.org/wmt26/model-compression.html`

[2] Model: `https://huggingface.co/alonsopg/wmt26-layeraware-native-mlp-q4`; experiments: `https://github.com/alonsopg/wmt26-layeraware-compression`.

[3] Organizer results and evaluator journal at revision `652d2bf`: `https://github.com/thammegowda/wmt26-model-compression/tree/652d2bf`.

System	Compressed scope	Representation and purpose
Organizer BF16	None	Uncompressed quality and performance reference.
Organizer global q4	All eligible linear modules	Organizer BitsAndBytes baseline: default FP4, no double quantization, FP32 compute.
Selected q3+Zstd	144 text-MLP projections	Transferred storage-codec method: decimal q3 rounding, FP16 storage, then Zstd level 3; decoded weights execute as BF16.
Layer-aware native q4	144 text-MLP projections	Submitted system: native NF4, double quantization, BF16 compute; all non-target modules remain BF16.

Table 1: Compared compression policies. The organizer global-q4 and submitted selective-q4 systems differ in both scope and 4-bit configuration; their comparison is a practical baseline comparison rather than a factorial isolation of layer selection.

model modules with the BitsAndBytes setting `load_in_4bit=True` and serializes the result. In the tested Transformers/BitsAndBytes versions, the stored configuration resolves to FP4 weights, FP32 compute, and no double quantization. We retain this baseline unchanged because it is the reference implementation provided to participants.

This detail matters for interpretation: the submitted system changes both the target-module policy and the low-bit configuration. Results against global q4 answer the operational question—how the submission compares with the organizer-provided q4 system—but do not by themselves identify layer selection as the sole cause of a quality difference.

3.2 Transferred storage-codec method

To connect this study to the previous selected-layer approach, we transfer its q3+Zstd policy to Gemma 3. The same 144 text-MLP matrices are rounded with `numcodecs.Quantize(digits=3, dtype=f4, astype=f2)` and compressed with Zstandard level 3 (Alted et al., 2016; Palomino, 2026). Quantization is lossy; Zstandard preserves the rounded representation exactly.

Storage is measured by actually encoding all selected tensors. The encoded selected-weight payload is 7.54 GiB; raw non-selected tensors and auxiliary files bring the estimated artifact to 14.46 GiB. For quality and generation, the exact q3 rounding is materialized into the BF16 backbone before inference. Consequently, post-load VRAM is meaningful, but recorded loading and generation omit Zstandard decompression. This system is an ablation, not a WMT26 submission candidate.

3.3 Layer-aware native MLP q4

Gemma 3 12B has 48 text-transformer blocks. For every block, we select the three gated-MLP projections: `gate_proj`, `up_proj`, and `down_proj`. The resulting 144 matrices contain 8.493 billion weight scalars. All attention projections, embeddings, normalization layers, the language-model head, and all vision-tower modules are placed on a skip list and retained in BF16.

Here, *layer-aware* means module-class-selective rather than block-adaptive: the same MLP-only policy is applied uniformly across all 48 text blocks.

The selected modules use BitsAndBytes NF4, double quantization, and BF16 computation. NF4 provides a 4-bit codebook designed for approximately normally distributed weights, while double quantization reduces overhead by quantizing the quantization constants (Dettmers et al., 2023). After model construction, we verify that the set of `Linear4bit` modules exactly equals the 144 intended targets before serializing the checkpoint with SafeTensors. The saved `layer_policy.json` records every target and preserved linear module.

Inspection of the serialized tensor payload gives the following size breakdown: 4.08 GiB for quantized text-MLP tensors, 4.22 GiB for other BF16 text tensors, 1.88 GiB for the tied token embedding/output head, and 0.78 GiB for the vision tower and multimodal projector; format and auxiliary-file overhead is about 33 MiB. Thus removing the unused vision path could save at most about 0.78 GiB before compatibility overhead, while quantizing the tied embedding/output matrix would target a larger 1.88 GiB component. We did not test either change.

Unlike the codec ablation, the submitted checkpoint retains its quantized representation when loaded. No custom decompression stage or reconstructed BF16 cache is required. The system passed the organizer sanity check, including the required line-count contract, and the final portable archive passed checksum and ZIP-integrity verification.

4 Results

4.1 Quality, size, memory, and speed

Table 2 reports the matched local comparison. The BF16 model gives the best absolute quality and fastest local generation, but has the largest native artifact and nearly the largest memory footprint. Both BitsAndBytes systems are slower than BF16 on the RTX A6000, illustrating that weight compression does not guarantee latency improvement on a particular GPU/kernel stack.

The submitted layer-aware system improves over organizer global q4 by 0.63 chrF and 0.56 BLEU. This recovery costs 3.21 GiB in artifact size and 2.60 GiB in peak VRAM, although local generation is 4.2% faster. Relative to BF16, the submitted artifact is 51.7% smaller and peak VRAM is 44.7% lower, with losses of 0.15 chrF and 0.28 BLEU. Its 2.175 versus 1.354 seconds/line is a 60.6% local slowdown, so BF16 dominates it on the local quality–speed plane. The submitted system instead occupies an intermediate quality–size and quality–memory operating point.

4.2 Paired significance

Table 3 gives paired-bootstrap differences for the submitted system. Both gains over global q4 are significant at $p < 0.05$. Against BF16, the chrF difference is not significant at that threshold, while the BLEU difference is. The comparison with the transferred codec finds no significant difference under either metric.

These tests measure reproducibility of score differences on the local set, not practical significance or blind-test generalization. They are especially useful for the global-q4 comparison: the 0.56 BLEU difference is small in absolute terms, but unlikely to result from segment sampling alone under this test.

4.3 Official blind-test results

The submission appears eighth among 20 constrained systems in the organizer table ordered by CometKiwi-XXL. Relative to global q4, it gains 0.0004 CometKiwi and improves MetricX by 0.0311, although the organizer reports no paired significance test and global q4 remains 3.45 GB smaller. Relative to BF16, it loses 0.0026 CometKiwi and 0.0108 MetricX while reducing size by 51.6%. H100 throughput is 0.1% below global q4 and 1.6% below BF16. Thus the large local A6000 slowdown relative to BF16 does not transfer to official H100 throughput, but compression provides no speed gain there either. Because the released `peak GB` measure is host RSS rather than GPU memory, the official results do not establish whether the local VRAM reduction transfers to H100.

4.4 Native q4 versus the previous codec

The transferred q3+Zstd method and native selective q4 have essentially the same local translation quality. Their deployment behavior is different. The codec artifact is 14.46 GiB and expands to 26.43 GiB peak VRAM after decoding, nearly identical to BF16's 26.46 GiB. Native q4 reduces artifact size by 24.0% and post-load peak VRAM by 44.6% relative to the codec. This is the main technical extension over the previous method: layer awareness is retained, but compression is represented in a form that survives model loading.

The codec's apparent generation speed should be treated cautiously. Its timed generation begins after rounded weights have been materialized; a complete codec runtime would also read and decompress 7.54 GiB of selected-weight payload. Native q4's timing includes its actual low-bit execution behavior, though not model loading, and is therefore the more deployment-relevant measurement of the two.

5 Submission and Reproducibility

The system was submitted as team `alonso`'s primary constrained English–Chinese entry. The public Hugging Face repository contains three SafeTensors shards, tokenizer/processor files, the quantization and layer-policy configurations, setup and run scripts, and pinned runtime requirements. The submitted repository revision begins `609c2568`. The organizer measured 11,804,443,105 on-disk bytes (11.8044 decimal GB, or 10.99 GiB); the experiment repository records the complete revision and final ZIP SHA-256 checksum. The package passed the organizer's first H100 sanity run without participant-side remediation.

Experiments use Transformers 4.57.6, BitsAndBytes 0.45.3, PyTorch 2.7.0, and SacreBLEU 2.5.1. The experiment repository includes model-policy

System	chrF	BLEU	Gen. s/line	Peak VRAM (GiB)	Artifact (GiB)
Organizer BF16	**28.45**	**31.65**	**1.354**	26.46	22.74
Organizer global q4	27.67	30.80	2.271	**12.04**	**7.78**
Selected q3+Zstd	28.26	31.39	1.358†	26.43	14.46
Layer-aware native q4	28.30	31.37	2.175	14.64	10.99

Table 2: Local results on 332 reference-backed WMT25 English–Chinese segments, batch size 8, one RTX A6000. Generation excludes processor/model loading. †The codec time also excludes Zstandard decompression and should not be interpreted as end-to-end codec latency.

Baseline	ΔchrF	p	ΔBLEU	p
Global q4	+0.63	.0003	+0.56	.0061
Selected codec	+0.05	.2561	−0.02	.3576
BF16	−0.15	.0897	−0.28	.0339

Table 3: Submitted native q4 minus each baseline, using 10,000 paired-bootstrap trials.

System	CK ↑	MX ↓	GB	char/s
Organizer BF16	0.7695	2.7728	24.4	881.1
Organizer global q4	0.7665	2.8147	8.4	867.9
Layer-aware native q4	0.7669	2.7836	11.8	866.9

Table 4: Official WMT26 English–Chinese blind results. CK is CometKiwi-XXL; MX is MetricX-24-XXL. Size uses decimal GB and throughput is source characters/s on the organizer H100 environment.

inventories, every prediction used in Tables 2 and 3, timing JSON, one-second GPU traces, exact artifact-size measurements, significance output, and the organizer sanity-check record. Large model files remain in the Hugging Face repository rather than GitHub.

6 Limitations

First, the controlled development comparison uses one RTX A6000 rather than the organizer's H100. Official H100 throughput changes the relative speed picture substantially, and the organizer did not release peak GPU-memory measurements, so local VRAM portability remains unresolved. The study also covers only English–Chinese; conclusions may not transfer to Czech–German, English–Egyptian Arabic, or other domains.

Second, paired significance is available only for local chrF and BLEU. The official CometKiwi and MetricX results agree directionally with the local comparison against global q4, but their margins are small and no paired significance or human evaluation is available. We therefore do not claim that the neural-metric differences are significant or human-perceptible.

Third, the practical organizer comparison is confounded: global q4 uses default FP4 without double quantization and FP32 compute, while the submitted system uses selective NF4 with double quantization and BF16 compute. A complete factorial ablation would compare global and selective policies under identical FP4 and NF4 configurations. The present results show that the submitted recipe outperforms the organizer recipe locally; they do not attribute the entire gain to layer selection alone.

Fourth, the codec storage figure combines an exactly measured encoded selected-weight payload with raw non-selected tensor bytes and auxiliary files. Its runtime emulation reproduces the codec's q3 rounding, but omits file I/O and Zstandard decompression. It therefore supports quality, storage, and post-decode-memory comparisons, not a full latency comparison.

Finally, the fixed policy preserves all vision modules even though the submitted direction is text-only, uses no permitted calibration data, and does not investigate per-layer sensitivity or calibration-aware mixed precision. Removing unused multimodal components or learning a finer-grained allocation could improve the Pareto trade-off, but would require additional compatibility testing.

7 Conclusion

This paper presented team alonso's WMT26 constrained English–Chinese submission, a mixed Gemma 3 12B checkpoint that natively quantizes only the text transformer's MLP gate, up, and down projections. On the local WMT25 reference set, the method significantly improves chrF and BLEU over the organizer global-q4 baseline while remaining larger in storage and VRAM. Official blind results show the same direction under CometKiwi-XXL and MetricX-24-XXL: quality is slightly better than global q4 and slightly below BF16, with nearly identical H100 throughput to both. The artifact is 48.4% of BF16 size.

The comparison with selected-layer q3+Zstd clarifies why native representation matters. The codec preserves quality but recovers BF16-like memory after decoding; native selective q4 retains low-bit storage during inference. These findings provide evidence that the selected-layer idea can transfer from speech translation to text MT in a more deployment-relevant form. The official evaluation supports a useful quality–size operating point, while leaving matched quantizer ablations and standardized GPU-memory measurement for future work.

Declaration of Generative Writing Assistance

OpenAI Codex was used to help organize experiment records and draft and edit this manuscript. The author reviewed the claims against the saved predictions, metrics, configurations, and runtime traces, edited the final text, and remains responsible for the paper's content. Codex was not used to generate evaluation references or alter system outputs.

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